

Experience foam for topical drug delivery



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Two major challenges in topical drug delivery are:

- Drug solubilization and adequate penetration of the *stratum corneum* to achieve effective therapeutic local delivery
- Patient adherence to treatment which is particularly low in topical drug administration

Functional excipients can help address these challenges by

- delivering effective drug solubilization and modulation of drug penetration through the *stratum corneum*
- being very safe and highly tolerable
- enabling the formulation of diverse topical bases with excellent texture and sensorial properties which improve patient experience and promote adherence

Depending on the active ingredient, disease/condition and target patient population, formulators can choose to develop different dosage forms such as traditional ointment, cream, lotion, microemulsion and gel or more innovative ones such as foams.

Pharmaceutical foams are becoming more popular probably due to a number of reasons including (among others) more topical medicines being switched to OTC and the need to innovate and differentiate 'brands', increasing awareness of the importance of formulating to deliver sensorial benefits and positive impact on patient experience and adherence.

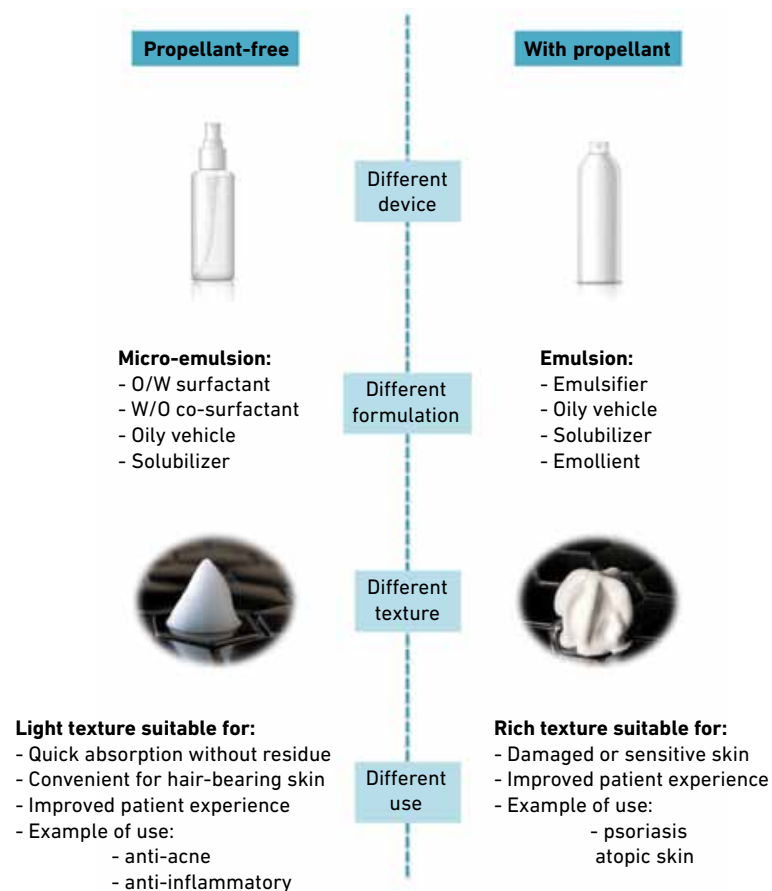
Numerous benefits have been reported for pharmaceutical foams (Kircik and Bikowski, 2012) such as:

- Ease of application, compared to ointment or lotion for example
- Ease of spreadability on application, which is:
 - o Suitable for diverse skin types, included hair-bearing skin
 - o Convenient to treat large zones
 - o Preferred for damaged skin for which low pressure and short application time are ideal
- Quick penetration and absorption
- May incorporate barrier repairing and emollient ingredients
- Lack of tackiness and shininess
- High level of patient satisfaction

Two distinct devices are available: with propellant gas (ie aerosol) or propellant-free, leading to two different formulation types and very different foam textures ([CALLOUT BOX 1](#)).

Gattefossé, as a leading provider of excipients for the pharmaceutical industry, manufactures solubilizers and texturizers for skin drug delivery. In this white paper we give formulation tips for propellant and propellant-free foams with guidelines on how to formulate and select excipients and case studies with model drugs.

CALLOUT BOX 1: Two strategies to develop pharmaceutical foams



From emulsion to Creamfoam using aerosol device



Today pharmaceutical foams are still marginal in the topical drug delivery market. Most of the prescription foams available contain corticosteroids, antibiotics, anti-acne and emollient agents. Foams are also well represented in OTC products such as Dicloream, Voltagan, Itamidol with sodium diclofenac.

Table 1: Examples of marketed foams (source www.rxlist.com, October 2016).

Brand name	Description
Ecoza	Econazole nitrate topical foam, 1%
Enstilar	Calcipotriene and betamethasone dipropionate foam, 0.005%/0.064%
Evoclin	Clindamycin phosphate
Extina	Ketoconazole foam, 2%
Fabior	Tazarotene
Hydro 35	Urea 35%, hydrating topical foam
Luxiq	Betamethasone valerate foam
Neosalus Foam	Hydrating topical foam
Olux-E	Clobetasol propionate foam
Proctofoam HC	Hydrocortisone and pramoxine rectal foam
Sorilux	Calcipotriene foam
Uceris	Budesonide rectal foam
Varithena	Polidocanol injectable foam
Verdeso	Desonide foam

With propellant devices, a light emulsion consisting of active ingredient, water, solubilizers, oily vehicles, emulsifiers and eventually emollient and stabilizing agents is used.

Developing a formulation step by step

1. Selection of solubilizers and emulsifiers

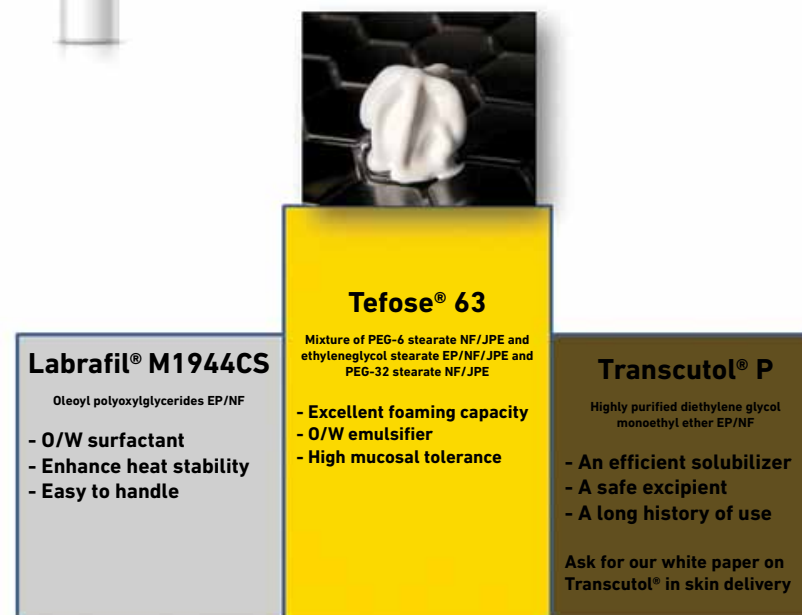
Solubility screening in different solubilizers is carried out to identify those that solubilize the therapeutic dose of drug – which is prerequisite for skin absorption. Then oily vehicles, emulsifiers and co-emulsifiers are selected depending on the chosen solubilizer(s).

Gattefossé winning trio of functional excipients for aerosol foam formulation are:

- Tefose®63 (PEG-6 stearate and glycol stearate and PEG-32 stearate) as emulsifier
- Labrafil® M 1944 CS (oleoyl polyoxyglycerides) as oily phase
- Transcutol® P (diethylene glycol monoethyl ether) as solubilizer



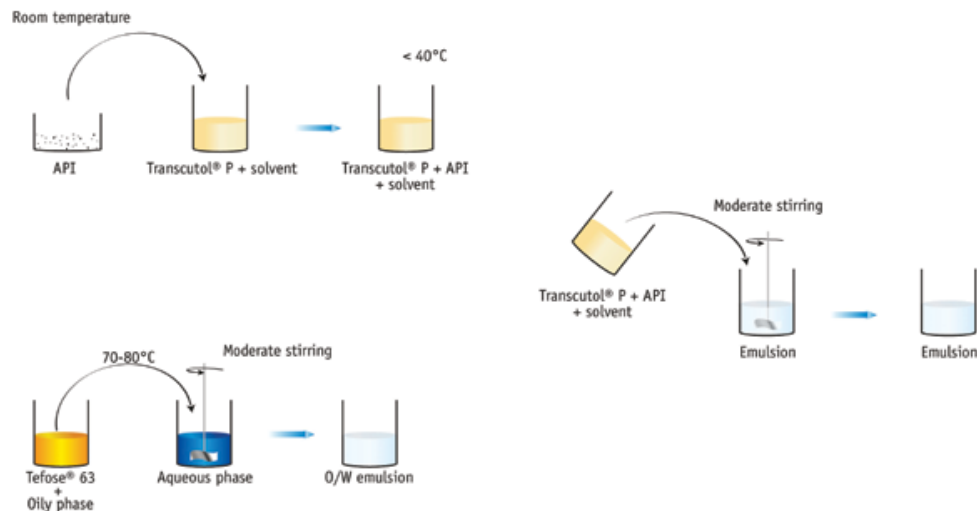
Winning trio for propellant foam



2. Preparing the emulsion

The emulsion is prepared using the following protocol:

- The API is dissolved at room temperature in Transcutol® and other solvents if required.
- Oily phase is prepared by mixing Tefose® 63 and Labrafil® M 1944 CS and heating the mixture at 70-80 °C.
- Aqueous phase is prepared by mixing water and preservative and heating the mixture at 70-80 °C.
- Incorporate the oily phase into the aqueous phase under moderate stirring to form an oil-in-water emulsion. Allow to cool down at 35 °C.
- Add the dissolved API into the emulsion under moderate stirring.



3. Evaluating foam characteristics

The formulation is assessed for:

- pH
- viscosity to enable correct flow through the device nozzle
- droplet size to assess emulsion homogeneity
- absence of drug crystal to ensure full solubilization throughout the process
- foamability and foam persistency

Sensory evaluation using a trained 'sensorial' panel is a useful tool as it enables the evaluation of a formulation and comparison with other formulations using key criteria which impact on patient experience, such as softness, thickness, spreadability and afterfeel. Sensory mapping therefore enables formulation optimization with improving patient experience as the objective

Gattefossé has an in-house sensory panel and expertise in sensorial analysis of topical products both in the cosmetic and pharmaceutical fields. This technique has been applied to characterize our foam formulations (see case study section) and compare their sensorial profile and advantages with conventional topical dosage forms.

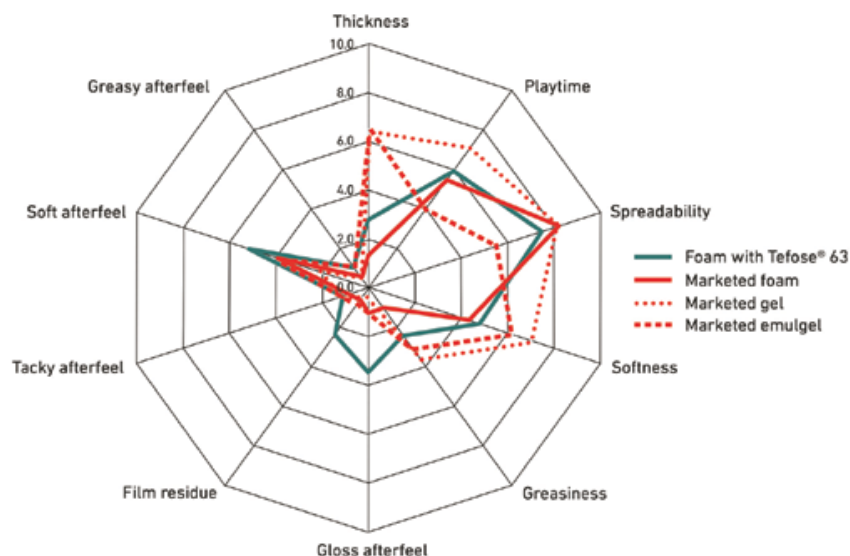


Case study #1: diclofenac sodium

A 1% diclofenac sodium optimized foam formulation was developed in our Application Laboratory (St Priest, France) with the following composition:

Ingredient	% W/W
Diclofenac sodium	1
Tefose® 63	8
Labrafil® M 1944 CS	4
Propylene glycol	5
Transcutol® P	5
Benzyl alcohol	1
Water	76

Sensorial analysis of this foam was compared to market references consisting of a foam, a gel and an emulgel.



The sensorial analysis highlights the benefits of using foam compared to conventional dosage forms: light texture, easy spreadability, softness during and after application, no tacky nor greasy afterfeel. Compared to the market foam, the foam with Tefose® 63 brings a higher comfort after application (gloss and afterfeel) with a better texture.

Case study #2: betamethasone dipropionate

A 0.064% betamethasone dipropionate optimized foam formulation was developed in our Application Laboratory (St Priest, France) with the following composition:

Ingredient	% W/W
Betamethasone dipropionate	0.05
Tefose® 63	6.00
Labrafil® M 1944 CS	3.00
Propylene glycol	5.00
Transcutol® P	2.00
Benzyl alcohol	1.00
Sodium hydroxide (10% sol.)	0.05
Water	82.90

ASK FOR

Our poster "Attractive emollient foams for pain treatments", Skin Forum, June 2016.

ASK FOR

Our poster "Converting light emulsion into creamy foam for psoriasis treatment. A case study with betamethasone dipropionate", AAPS, November 2016.

From microemulsion to Lightfoam using propellant-free device

From microemulsion to Lightfoam using propellant-free device



Propellant-free foams are widely used in cosmetics but under exploited in pharma. Their main interest lays in their simplicity, both in formulation and device.

The device contains a simple microemulsion combining the API, a surfactant, a co-surfactant and an oily vehicle. The surfactant exerts a dual role as foaming agent and solubilizer.

The benefits of microemulsion are well known:

- Easy to formulate
- Simple mixing process
- Thermodynamically stable systems
- Increased cutaneous drug delivery (Kreilgaard, 2002)

Putting the microemulsion into a propellant-free foam device enables the transformation of a liquid product into a light foam which is easy to spread and remains at the site of application.

Developing a formulation step by step

1. Screening and selection of excipients for drug solubility

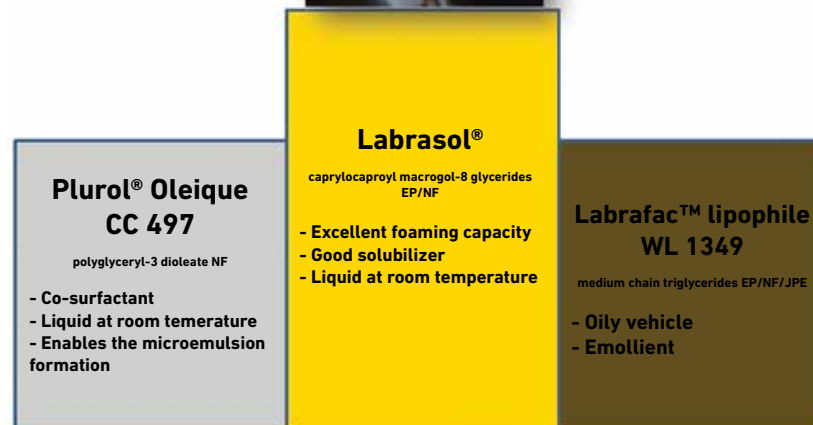
Solubility screening of surfactants, co-surfactants, oily vehicles and solubilizers is carried out to highlight the best candidates.

Gattefossé winning trio of functional excipients for propellant-free foam formulation are:

- Labrasol® (caprylocaproyl macrogol-8 glycerides) as surfactant and solubilizer
- Plurol® Oleique CC 497 (polyglyceryl-3 dioleate) as co-surfactant
- Labrafac™ lipophile WL 1349 (medium chain triglycerides) as oily phase and emollient

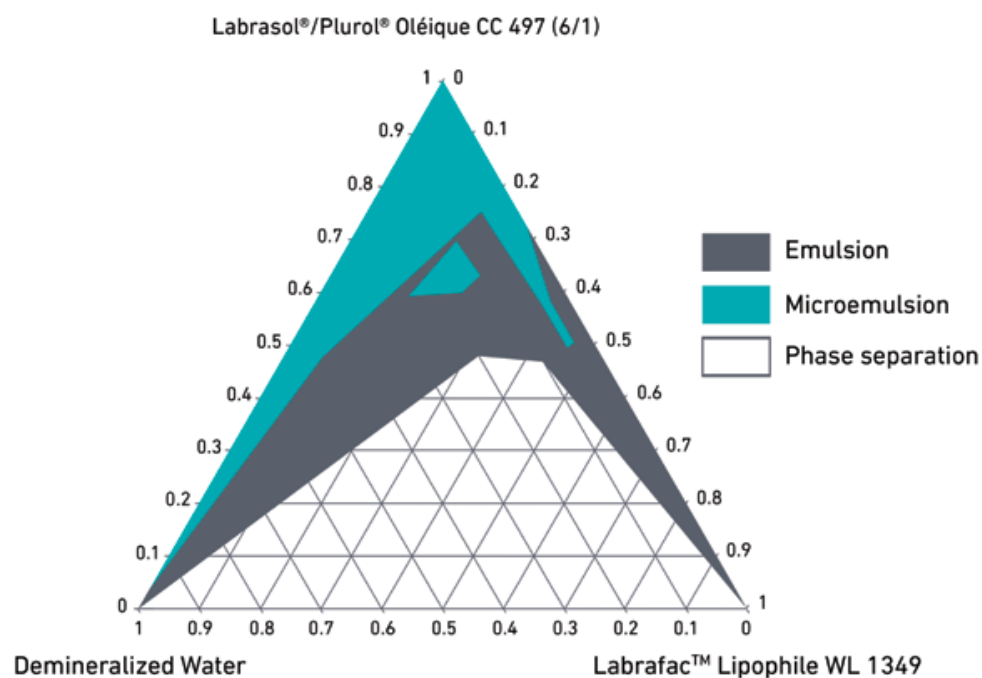


Winning trio for propellant-free foam



2. Construction of pseudo-ternary diagrams to identify micro-emulsions areas

Once the excipients have been selected a pseudo-ternary diagram is constructed to determine the relative concentrations and ratios required to obtain a microemulsion when in contact with aqueous solution i.e. (in green zone on the graph below).



3. Addition of API

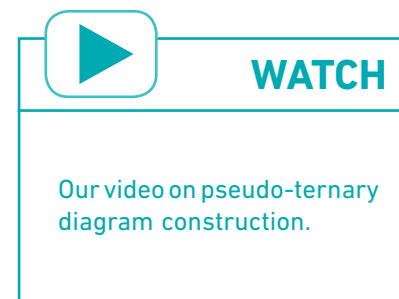
API is added once the excipient quantities have been determined. It is important to check that the addition of the drug has not displaced the microemulsion area. This is done by visual assessment and particle size evaluation.

4. Evaluating foam capacity and characteristics

The formulation is assessed for:

- pH
- viscosity to ensure correct flow
- absence of drug crystal to ensure full solubilization throughout the process
- foamability and foam persistency which is a visual assessment of foam quality after three minutes

A sensory evaluation can also be performed to identify the key sensorial properties of the foam.



Case study #3: diclofenac sodium

Using the method described above (ie solubility screening and ternary diagram construction), simple microemulsions containing 1 or 3% sodium diclofenac were produced (see table of composition). The foam capacity and persistency were very good.

Ingredient	1% diclofenac formula	3% diclofenac formula
	% W/W	% W/W
Diclofenac sodium	1.0	3.0
Labrasol®	18.0	24.0
Plurol® Oleique CC497	3.0	4.0
Labrafac™ Lipophile WL 1349	0.5	0.5
Water	77.5	68.5



ASK FOR

Our poster “Pharmaceutical foams with optimized microemulsion and propellant-free pump”, PBP, April 2016.



WATCH

Our talking poster on pharmaceutical foams.

Case study #4: lidocaine base

A 4% lidocaine base foam formulation was successfully developed following this protocol, with good foamability and persistency.

Ingredient	% W/W
Lidocaine base	4.0
Labrasol®	24.0
Plurol® Oleique CC497	4.0
Labrafac™ Lipophile WL 1349	0.5
Benzyl alcohol	1.0
Water	66.5



ASK FOR

Our poster “Converting simple microemulsion into attractive foam for topical pain treatment. A case study with lidocaine base”, AAPS, November 2016.

Want to experience the foam?

There is increasing interest in pharmaceutical foams due to their relative novelty and advantages for patients.

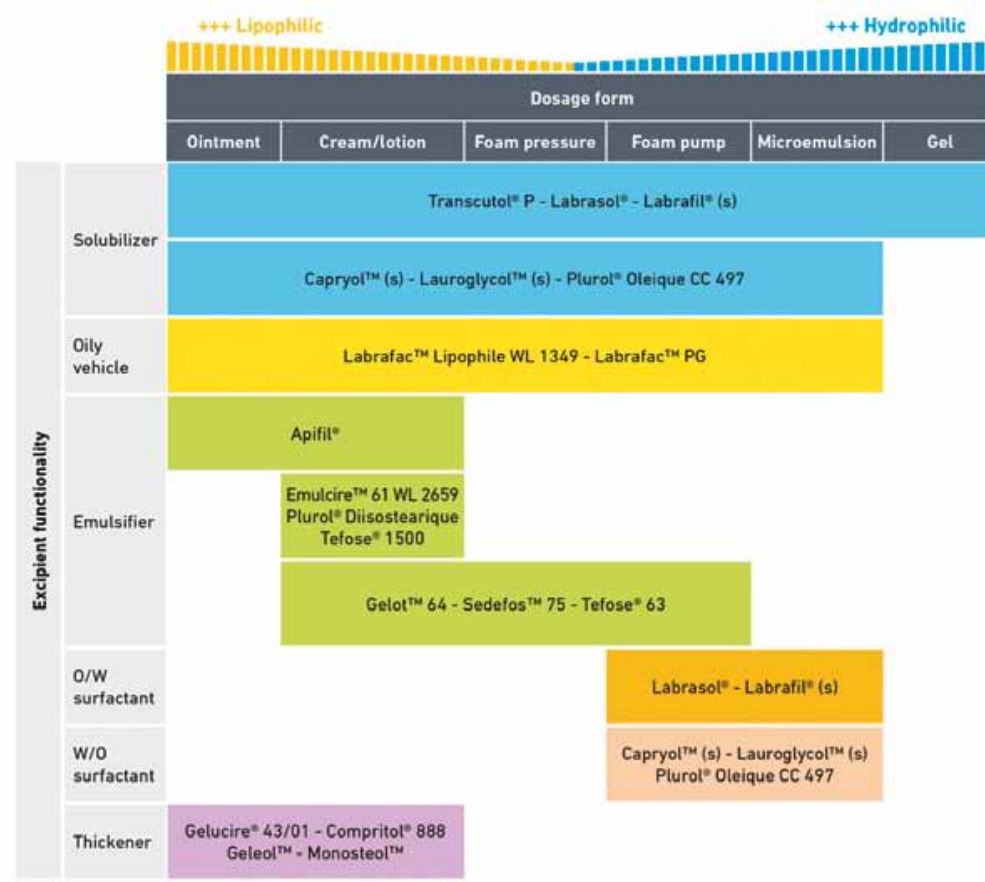
Aerosol foams are already available for certain topical treatments but the dosage form is not widely used possibly due to cost/benefit and environmental issues associated with aerosols.

Propellant-free foams are under-exploited whereas they are of high interest due to the possibility to exploit the advantages of a microemulsion formulation in a propellant-free device.

Gattefossé has a range of specialty excipients for aerosol and propellant-free foam devices and has expertise in topical formulations with application laboratories around the world.

The excipients highlighted in this paper are "our best of" among a full range available.

Please do not hesitate to contact us for further information at infopharma@gattefosse.com



www.gattefosse.com

Bibliography

Kircik and Bilowski, 2012, The science of topical foam formulations, Supplement to Practical Dermatology, January 2012.

Kreilgaard, M. (2002). Influence of microemulsions on cutaneous drug delivery. Advanced Drug Delivery Reviews, 54, S77-S98.

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